# Hidden hills in the old basket model, and why the mixture form has none

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**Abstract**

Both the old Version-4 basket model and its mixture form are built from the same exact law of a basket given a latent taste $z$. They differ only in the weight each taste receives. In the old model that weight is the Gaussian density times $Z_z$, the total score of all baskets the taste allows. Because $Z_z$ grows fastest for tastes that make large baskets attractive, the weight can have several hills, one of them far from every real basket; training and evaluation that look only near the data never see it. In the mixture form the weight is the Gaussian density alone, and the posterior of the taste given any basket has exactly one hill. A worked example with 20 products shows both effects with numbers that can be checked by hand.

## 1 Notation

- One trip is fixed throughout. $\mathcal{J}$ is the set of products offered on it. Each product $j \in \mathcal{J}$ belongs to one category $c(j)$, and $\mathcal{J}_c = \{j : c(j) = c\}$.
- A *basket* is a non-empty set $S \subseteq \mathcal{J}$. For a basket, $n(S) = |S|$ is its size, $n_c(S) = |S \cap \mathcal{J}_c|$ the number of its products in category $c$, and $m(S) = \#\{c : n_c(S) > 0\}$ the number of categories present.
- $\mathcal{S}$ is the set of allowed baskets: $1 \le n(S) \le 120$ and $n_c(S) \le 40$ for every $c$. Every sum over baskets below runs over $\mathcal{S}$.
- The taste is a vector $z \in \mathbb{R}^r$ ($r = 16$ on the real data, $r = 1$ in the examples). $\mathcal{N}(z; 0, I) = (2\pi)^{-r/2} e^{-\|z\|^2/2}$ is the standard Gaussian density, the "bell curve".
- $(a)_+ = \max(a, 0)$ and $\binom{k}{2} = k(k-1)/2$.

**Parameters.** Each product has a context utility $b_j \in \mathbb{R}$ (price, promotion, season, household taste, history; a fixed number for this trip) and a taste loading $\phi_j \in \mathbb{R}^r$. Each category has a taste loading $\psi_c \in \mathbb{R}^r$ for its extra products, a pair penalty $\rho_c$ and count penalties $f_c(k)$ (with $f_c(0) = f_c(1) = 0$). $\rho_0(m)$ is the size penalty on the number of categories present.

## 2 The common building block: the basket law given a taste

**Definition 1** (score and total score)**.** *For a taste $z$ and a basket $S$, the* score *of $S$ is $e^{E_z(S)}$ with*

$$E_z(S) = \sum_{j \in S} u_j(z) - \sum_c h_c\big(n_c(S), z\big) - \rho_0\big(m(S)\big), \tag{1}$$

$$u_j(z) = b_j - \tfrac{1}{2}\|\phi_j\|^2 + z \cdot \phi_j,$$

$$h_c(k, z) = \rho_c \tbinom{k}{2} + f_c(k) + (k-1)_+\big(z \cdot \psi_c - \tfrac{1}{2}\|\psi_c\|^2\big).$$

*The* total score *of taste $z$ is the sum of the scores of all allowed baskets,*

$$Z_z = \sum_{S \in \mathcal{S}} e^{E_z(S)}. \tag{2}$$

**Definition 2** (basket law given a taste)**.**

$$p(S \mid z) = \frac{e^{E_z(S)}}{Z_z}. \tag{3}$$

A basket's probability is its share of the total score; dividing by $Z_z$ makes the probabilities of all baskets add up to one. Both models below use exactly this law.

**The basket field.** Collecting the terms of (1) that contain $z$ gives

$$E_z(S) = a(S) + z \cdot s(S), \qquad s(S) = \sum_{j \in S} \phi_j - \sum_c (n_c(S) - 1)_+ \, \psi_c, \tag{4}$$

where $a(S)$ collects every term of (1) without $z$. The vector $s(S) \in \mathbb{R}^r$, the *basket field*, is the direction in taste space that basket $S$ points in.

## 2.1 Example 1: three products

Take $\mathcal{J} = \{A, B, C\}$, no penalties ($\rho_c = f_c = \rho_0 = 0$, $\psi_c = 0$), and a taste $z$ under which $u_A(z) = 1$, $u_B(z) = 0$, $u_C(z) = -1$. Then $e^{E_z(S)} = \exp \sum_{j \in S} u_j(z)$ and

| $S$ | $\{A\}$ | $\{B\}$ | $\{C\}$ | $\{A,B\}$ | $\{A,C\}$ | $\{B,C\}$ | $\{A,B,C\}$ | $Z_z$ |
|---|---|---|---|---|---|---|---|---|
| $e^{E_z(S)}$ | 2.72 | 1.00 | 0.37 | 2.72 | 1.00 | 0.37 | 1.00 | 9.17 |

so $p(\{A\} \mid z) = 2.72/9.17 = 0.30$ and $p(\{A, B, C\} \mid z) = 1.00/9.17 = 0.11$.

Now take a taste $z'$ that raises every utility by 2, so $u_A = 3$, $u_B = 2$, $u_C = 1$:

| $S$ | $\{A\}$ | $\{B\}$ | $\{C\}$ | $\{A,B\}$ | $\{A,C\}$ | $\{B,C\}$ | $\{A,B,C\}$ | $Z_{z'}$ |
|---|---|---|---|---|---|---|---|---|
| $e^{E_{z'}(S)}$ | 20.1 | 7.4 | 2.7 | 148.4 | 54.6 | 20.1 | 403.4 | 656.7 |

Raising each utility by 2 multiplies the score of a basket of $n$ products by $e^{2n}$. The total score grows about 72-fold, almost all of it from the largest basket, whose probability rises to $403.4/656.7 = 0.61$. With thousands of products and baskets of up to 120, a taste that raises many utilities makes $Z_z$ astronomically large, and nearly all of it comes from huge baskets.

## 2.2 How $Z_z$ is computed and how it changes with $z$

$\mathcal{S}$ is far too large to list, but $Z_z$ is computed exactly. For each category $c$ and each count $k$, the total score of choosing $k$ of its products is $G_c(k) = e^{-h_c(k,z)} \sum_{A \subseteq \mathcal{J}_c, |A| = k} \prod_{j \in A} e^{u_j(z)}$, by a recursion over the products of $c$. The categories are then combined in a table indexed by the basket size $n$ and the number of categories $m$, and $Z_z = \sum_{n,m} (\text{table entry}) \, e^{-\rho_0(m)}$.

For the conditional mean and covariance of the basket field, write $\mathbb{E}[s \mid z] = \sum_S p(S \mid z) \, s(S)$ and $\mathrm{Cov}[s \mid z] = \sum_S p(S \mid z) \, (s(S) - \mathbb{E}[s \mid z])(s(S) - \mathbb{E}[s \mid z])^\top$.

**Lemma 1.** $\nabla_z \log Z_z = \mathbb{E}[s \mid z]$ *and* $\nabla_z^2 \log Z_z = \mathrm{Cov}[s \mid z]$. *In particular* $\log Z_z$ *is convex in* $z$.

*Proof.* By (4), $\log Z_z = \log \sum_S e^{a(S) + z \cdot s(S)}$. Differentiating once gives $\sum_S e^{a(S) + z \cdot s(S)} s(S) / Z_z = \mathbb{E}[s \mid z]$; differentiating again gives the covariance, which is positive semi-definite. □

In words: moving the taste in a direction $v$ raises $\log Z_z$ at the rate $v \cdot \mathbb{E}[s \mid z]$, the expected basket field along $v$, and that rate itself only increases as $z$ moves further along $v$.

## 3 The old model and its taste weight

**Definition 3** (old model)**.**

$$p_{\text{old}}(S) = \frac{e^{a(S)+\frac{1}{2}\|s(S)\|^2}}{Z_{\text{old}}}, \qquad Z_{\text{old}} = \sum_{S\in\mathcal{S}} e^{a(S)+\frac{1}{2}\|s(S)\|^2}. \tag{5}$$

The term $\frac{1}{2}\|s(S)\|^2$ is the "goes-together" bonus: expanding it, it contains $\phi_i \cdot \phi_j$ for every pair of products in $S$, so products whose loadings point the same way reward each other. For a basket of $n$ products with similar loadings it grows like $n^2$, while the penalties in $a(S)$ grow like $n$.

**Lemma 2** (Gaussian identity)**.** *For every $s \in \mathbb{R}^r$, $e^{\frac{1}{2}\|s\|^2} = \int_{\mathbb{R}^r} \mathcal{N}(z;0,I)\, e^{z\cdot s}\, dz$.*

*Proof.* $-\frac{1}{2}\|z\|^2 + z\cdot s = -\frac{1}{2}\|z-s\|^2 + \frac{1}{2}\|s\|^2$, and $\mathcal{N}(\cdot\,;s,I)$ integrates to one. □

**Proposition 1** (the old model as a weighted mixture)**.**

$$p_{\text{old}}(S) = \int_{\mathbb{R}^r} \pi_{\text{old}}(z)\, p(S \mid z)\, dz, \qquad \pi_{\text{old}}(z) = \frac{\mathcal{N}(z;0,I)\, Z_z}{Z_{\text{old}}}. \tag{6}$$

*Proof.* By Lemma 2 and (4), $e^{a(S)+\frac{1}{2}\|s(S)\|^2} = \int \mathcal{N}(z;0,I)\, e^{E_z(S)}\, dz = \int \mathcal{N}(z;0,I)\, Z_z\, p(S \mid z)\, dz$. Dividing by $Z_{\text{old}}$ gives (6). Summing over $S$ shows that $\pi_{\text{old}}$ integrates to one. □

So the old model also picks a taste and then a basket, but a taste is weighted by the bell curve *multiplied by its total score* $Z_z$. A taste that makes many high-scoring baskets possible gains weight.

**Proposition 2** (slope and curvature of the old weight)**.**

$$\nabla_z \log \pi_{\text{old}}(z) = -z + \mathbb{E}[s \mid z], \qquad \nabla_z^2 \log \pi_{\text{old}}(z) = -I + \text{Cov}[s \mid z].$$

*Proof.* $\log \pi_{\text{old}}(z) = -\frac{1}{2}\|z\|^2 + \log Z_z + \text{const}$; apply Lemma 1. □

Two forces act on the slope: $-z$ pulls back towards zero, and $\mathbb{E}[s \mid z]$ pushes outward, harder the further out $z$ is (Lemma 1). Where $\text{Cov}[s \mid z]$ exceeds $I$ the curvature is positive, so $\log \pi_{\text{old}}$ is not concave and can have several local maxima, the *hills*. Each hill satisfies $z = \mathbb{E}[s \mid z]$.

### 3.1 Example 2: twenty products and a hidden hill

Take one taste dimension ($r = 1$), 20 products in 20 different categories, all with $b_j = -4$ and $\phi_j = 1$, and no penalties or $\psi$. Then $u_j(z) = -4.5 + z$ for every product, $s(S) = n(S)$, and

$$Z_z = \left(1 + e^{-4.5+z}\right)^{20} - 1, \qquad \mathbb{E}[n \mid z] = \frac{20\,\sigma(-4.5+z)\,(1+e^{-4.5+z})^{20}}{Z_z}, \qquad \sigma(x) = \frac{1}{1+e^{-x}},$$

where the $-1$ removes the empty basket. The slope of Proposition 2 is $-z + \mathbb{E}[n \mid z]$. Its zeros, computed numerically:

| $z$ | $\mathbb{E}[n \mid z]$ | $\log \mathcal{N}(z;0,1) Z_z + \frac{1}{2}\log 2\pi$ | |
|---|---|---|---|
| 1.56 | 1.56 | $-0.63$ | hill 1: near the data |
| 2.36 | 2.36 | $-0.68$ | valley |
| 20.0 | 20.0 | $+110.0$ | hill 2: all 20 products bought |

Hill 2 is about $e^{110.6}$ times higher than hill 1. Integrating (6) numerically, the region $z < 5$ holds a fraction $10^{-46}$ of the old model's probability, so under $p_{\text{old}}$ essentially every basket contains all 20 products.

The same follows directly from (5). The full basket has $a(S) = 20 \times (-4.5) = -90$ and bonus $\frac{1}{2} \cdot 20^2 = 200$, energy 110; a single product has energy $-4.5 + \frac{1}{2} = -4$. The pair bonus wins.

**Why fitting and evaluation miss hill 2.** The gradient of the old log-likelihood in any parameter $\vartheta$ is "observed minus expected":

$$\nabla_\vartheta \log p_{\text{old}}(S) = \nabla_\vartheta \big[a(S) + \tfrac{1}{2}\|s(S)\|^2\big] - \mathbb{E}_{S' \sim p_{\text{old}}} \nabla_\vartheta \big[a(S') + \tfrac{1}{2}\|s(S')\|^2\big].$$

The expectation is over $p_{\text{old}}$, and by (6) over $\pi_{\text{old}}$. If real baskets hold one or two products, chains started from them stay around hill 1, and so does any proposal built from them. The estimate of the expectation then covers hill 1 only, nothing pushes hill 2 down, and an estimate of $Z_{\text{old}}$ that misses hill 2 makes the reported likelihood too high. On the Dunnhumby refit this hill was the region of 120-product baskets.

# 4 The mixture model

**Definition 4** (mixture model)**.**

$$p(S) = \int_{\mathbb{R}^r} \mathcal{N}(z; 0, I)\, p(S \mid z)\, dz. \tag{7}$$

The law given $z$ is the same (3); the taste weight is $\mathcal{N}(z; 0, I)$ alone. $Z_z$ appears only as the divisor inside $p(S \mid z)$, which makes each taste's probabilities add up to one. A taste that allows many high-scoring baskets gains no weight; it shares its fixed weight among more baskets. $p(S)$ sums to one because $\sum_S p(S \mid z) = 1$ for every $z$.

**Generation.** Draw $z \sim \mathcal{N}(0, I)$, then $S \sim p(\cdot \mid z)$ exactly by the dynamic program of Section 2.2 (size and categories first, then the count in each category, then the products). No search is involved, so nothing can be missed.

**Example 2 again.** Under (7) the taste $z = 20$ has density $\propto e^{-200}$. Typical tastes lie in $[-2, 2]$, where each product is bought with probability $\sigma(-4.5 + z) \in [0.0015, 0.076]$ and $\mathbb{E}[n \mid z] \in [1.01, 1.91]$. The model's mean basket size is $\int \mathcal{N}(z; 0, 1)\mathbb{E}[n \mid z]\, dz = 1.19$, and the probability of the full 20-product basket is $2.3 \times 10^{-10} \approx e^{-22.2}$.

## 4.1 The posterior of the taste has one hill

Scoring an observed basket $S$ requires the tastes that explain it. By Bayes' rule their weight is the posterior

$$p(z \mid S) = \frac{\mathcal{N}(z; 0, I)\, p(S \mid z)}{p(S)}, \qquad \log p(z \mid S) = -\tfrac{1}{2}\|z\|^2 + z \cdot s(S) - \log Z_z + \text{const}, \tag{8}$$

using (4); the constant does not depend on $z$.

**Proposition 3** (one hill)**.**

$$\nabla_z \log p(z \mid S) = -z + s(S) - \mathbb{E}[s \mid z], \qquad \nabla_z^2 \log p(z \mid S) = -I - \text{Cov}[s \mid z] \preceq -I.$$

*Hence* $\log p(z \mid S)$ *is strictly concave, has exactly one maximum* $\hat{z}$*, and* $\log p(z \mid S) \le \log p(\hat{z} \mid S) - \frac{1}{2}\|z - \hat{z}\|^2$ *for every* $z$*.*

*Proof.* Differentiate (8) with Lemma 1. A covariance is positive semi-definite, so the Hessian is at most $-I$ in every direction. A function whose Hessian is at most $-I$ everywhere has a unique maximum, and the quadratic bound follows from Taylor's theorem. □

Compare with Proposition 2. The term $\mathbb{E}[s \mid z]$ that pushed the old weight outward appears here with a minus sign: it now works with the bell curve instead of against it. The basket enters only through the constant vector $s(S)$, which shifts the hill but cannot create a second one. This holds for every basket, every parameter value and every dimension $r$.

**Example 2 again.** For $S = \{\text{one product}\}$ the slope is $-z + 1 - \mathbb{E}[n \mid z]$. Both $-z$ and $-\mathbb{E}[n \mid z]$ decrease in $z$, so the slope crosses zero once, at $\hat{z} = -0.10$.

### 4.2 Consequences

- *Scoring.* $p(S)$ is estimated by importance sampling around the single hill $\hat{z}$, found by gradient ascent, which always reaches it. There is no second hill to miss.
- *Training.* The gradient is $\sum_k \bar{w}_k \big(\nabla E_{z_k}(S) - \mathbb{E}_{S' \sim p(\cdot \mid z_k)} \nabla E_{z_k}(S')\big)$, with tastes $z_k$ drawn around $\hat{z}$ and normalised importance weights $\bar{w}_k$. The inner expectation is exact for each $z_k$ (it is the gradient of $\log Z_{z_k}$), so no region of baskets is left unvisited.
- *Generation* is exact (above).

| | old model | mixture model |
|---|---|---|
| law given $z$ | $p(S \mid z)$, (3) | the same |
| weight of taste $z$ | $\mathcal{N}(z; 0, I)\, Z_z / Z_{\text{old}}$ | $\mathcal{N}(z; 0, I)$ |
| slope of the log weight that matters | $-z + \mathbb{E}[s \mid z]$ | $-z + s(S) - \mathbb{E}[s \mid z]$ |
| curvature | $-I + \text{Cov}[s \mid z]$ | $-I - \text{Cov}[s \mid z]$ |
| number of hills | one or more | exactly one |
| drawing a basket | Markov chain | exact, $z$ then $S \mid z$ |

## 5 What changes and what does not

- *It is a different model.* The two forms share every parameter but define different laws, so a fit of one is not a fit of the other.
- *Prices and tastes.* In the old model a price change alters $Z_z$ and so shifts which tastes are common. In the mixture the taste distribution is fixed by $\mathcal{N}(0, I)$; prices change only what each taste buys. This is an assumption, suited to short-run price and promotion questions.
- *What it does not fix.* The mixture removes hidden hills; it does not by itself make training correct. In the full real fit the held-out log-likelihood on all 1,024 validation trips fell from $-38.10$ at step 240 to $-38.76$ at the end (paired difference $-0.66 \pm 0.08$), and the drop is real at 2,048 importance draws. That is a training problem, to be analysed separately.